

Use Case 1: Initialize Launch**Scope:** Launch Mission**Primary Actor:** Flight Director**Related Use Cases:** *Use Case 1.1 Initialize Rocket**Use Case 1.2 Initialize Launch Platform***Stake Holders & Interests:**

- Flight Director: Wants to set up a Launch by Setting all the Launch Parameters and Sub-Systems including
 - Rocket
 - Capsule
 - Launching Platform
 - Launching Mechanism(s)
- Mission Planner: Wants the Mission Started with the appropriate Mission Parameters loaded
- Engineers/Technicians: Wants to monitor the entire launch to ensure success

Pre-Conditions: The Mission System is running and ready to monitor all the appropriate Systems and Sub-Systems.**Post-Conditions:** The Launching Mission System is Initialized with the proper Launching Parameters and Monitoring all Launch Systems and Sub-Systems**Flow of Events:**

Flight Director	System
1. Enter the Mission Parameters File	
	2. Opens Mission Parameters File
	3. Reads the Parameters File
	4. Initialize the Rocket and its Systems by sending the Mission Parameters File to the Rocket Sub-System
	5. Alerts Flight Director of the Rocket Initialization Status
	6. Initialize the Launch Platform and its Systems by sending the Mission Parameters File to the Launch Platform and its Systems
	7. Alerts Flight Director of the Launch Platform Initialization Status

Extensions (or Alternative Flows):

- 2a. If there is an error in opening the Parameters file, then the System alerts the Flight Director of the error: indicating unable to open the Parameters file. The Flight Director either chooses another Parameters file or exits (aborts) the Mission.

- 3a. If the System cannot read the Parameters file, then the System alerts the Flight Director of the error: indicating unable to read the Parameters file. The Flight Director either chooses another Parameters file or exits (aborts) the Mission.
- 4a. If the Rocket and/or Sub-Systems is unable to open the Parameters file, the System alerts the Flight Director of the Error: indicating unable to open the file. The Flight Director either chooses another Parameters file or exits (aborts) the Mission
- 4b. If the Rocket and/or Sub-Systems are unable to read the Parameters file, then the System alerts the Flight Director of the Error: indicating unable to read the file. The Flight Director either chooses another Parameters file or exits (aborts) the Mission.
- 6a. If the Launch Platform and/or Sub-Systems are unable to open the Parameters file, then the System alerts the Flight Director of the Error: indicating unable to open the file. The Flight Director chooses course of action.
- 6b. if the Launch Platform and/or Sub-systems are unable to read the Parameters file, then the System alerts the Flight Director of the Error: indicating unable to read the file. The Flight Director chooses next course of action.
- 7a+ After the System is initialized with data The System periodically monitors the Rocket and returns the Current Conditions including any Errors
- 7b+ After the System is initialized with the data the System periodically monitors Launch Platform and returns the Current Condition including any Errors

Special Requirements:

- None

Technology & Variations List:

- If the System queries the Sub-Systems, error state and data shall be included.
- If the Sub-Systems alert, the Sub-Systems either alert in a separate error message or the same update message.

Frequency of Occurrence:

- Every time a planned Mission is exercised or simulated.

Open Issues:

- During Initialization, how is System Monitoring accomplished?

- Periodic monitoring of the Subsystems
- Subsystems periodic alert
- Is there the ability to implement both types of monitoring?